Bone structure and classification

BONE STRUCTURE AND CLASSIFICATION (sample notes)

Gross structure
- Compact bone: dense outer layer; osteons (Haversian systems) run parallel to the long axis.
- Spongy (cancellous) bone: trabeculae enclosing red marrow in many adult bones.
- Periosteum: fibrous membrane covering the outer surface of bone; essential for growth, repair and blood supply.
- Endosteum: lines the medullary cavity.
- Medullary cavity: houses yellow marrow (fat) in adult long bones.

Classification by shape
- Long bones (femur, humerus): longer than wide.
- Short bones (carpals, tarsals): roughly cube shaped.
- Flat bones (skull vault, sternum, scapula): thin and broad.
- Irregular bones (vertebrae): complex shapes.

Bone cells
- Osteoblasts build bone matrix; osteocytes maintain it; osteoclasts resorb bone.

Clinical link: osteoporosis reflects an imbalance where osteoclast activity outpaces osteoblast activity, reducing bone density and increasing fracture risk.